TEXORA · LECTURE SHEET

Woven Fabric Structure: Introduction, Classification and Representation

A woven fabric is fully described by the order in which its two yarn systems interlace. This sheet fixes the vocabulary of the whole course — warp and weft, repeat, and the four paper records (design, draft, peg plan, denting plan) from which every structure in the following topics is built and read.

TE-3205 · FABRIC STRUCTURE AND DESIGN · TOPIC 1 · LEVEL 3 TERM 2

1. What a woven fabric is

A **woven fabric** is a planar structure made by interlacing two sets of yarns at right angles on the loom. The lengthwise set is the **warp**: many parallel ends, wound on the warp beam, each end controlled through a heald (harness) so it can be raised or lowered. The crosswise set is the **weft**: a single yarn (a **pick**) inserted across the raised/lowered warp sheet (the **shed**) once per loom cycle, then beaten into the cloth fell by the reed. Wherever a warp end is **raised** for a pick, the pick passes *under* that end and the end shows on the face; wherever the end is **lowered**, the pick passes over it and the weft shows. The pattern of these decisions — which ends are up for which picks — *is* the weave, and everything else in this course is notation and consequences of that pattern.

Woven structures contrast with the other fabric families: **weft/warp knitted** fabrics (one yarn system forming intermeshed loops — TE-3205 Topics 9–12 of this course), **nonwovens** (fibres bonded directly, no yarn order to design), and **braids/laces** (yarns interlaced diagonally or twisted). Only weaving starts from two independent, pre-planned yarn systems, which is why its design can be written down completely on squared paper before any yarn is woven.

2. First vocabulary: density, crimp, repeat

- **Ends and picks.** Warp density is counted as **ends per unit length** (per cm in this course; per inch, EPI, in mill practice), weft density as **picks per unit length** (PPI). TE-3206 measures both from real samples.
- **Crimp.** A yarn in fabric does not run straight: it bends around the crossing system. Crimp % = (straightened yarn length − fabric length) ÷ fabric length × 100. The more interlacings per unit length, the more crimp and the firmer the cloth; plain weave (Topic 2) crimps both systems most.

- **Repeat.** The smallest rectangle of ends × picks after which the interlacing pattern repeats exactly. Repeat size is quoted as *ends* × *picks* (e.g. 4 × 4). A design is drawn for one repeat, sometimes two side by side to show the join.
- **Float.** A length of yarn lying over (or under) two or more threads of the crossing system without interlacing. Floats decide lustre, smoothness, cover and snagging: satin (Topic 4) is the extreme case.

3. Classification of woven structures

Structures are classified in three independent ways, and a complete description uses all three.

BASIS	GROUPS
Elementary weave	**Plain, twill, satin/sateen** — the three fundamental interlacings from which all others derive (Topics 2–4).
Derived weave	Systematic modifications of the fundamentals: rib and hopsack from plain; pointed, broken, combined and diamond twills from twill; rearranged and crepe structures; colour-and-weave effects (Topics 2, 3, 5, 6).
Compound (multi-system) structure	More than two yarn systems: **double cloth** (2 warps + 2 wefts, Topic 7), **pile fabrics** (an extra warp or weft forms loops or cut ends, Topic 8), backed cloths, extra-figured designs.

By layer, fabrics are **single-layer** (one warp, one weft) or **multi-layer** (double and treble cloths). By surface they are **flat** or **pile**. Note carefully: knitted structures (Topics 9–12) are classified by a different scheme — by loop formation, not interlacing — and must never be described in weave vocabulary.

4. Representation on design (point) paper

The universal record is a grid of squares: **each vertical column of squares is one warp end; each horizontal row is one pick; a filled (marked) square means that end is raised over that pick**, so the warp shows on the face there; a blank square means the end is down and the weft shows. By universal convention the design is read with **pick 1 at the bottom** — the first pick woven is the lowest row — and end 1 at the left. Two repeats are often drawn to prove the pattern joins correctly.

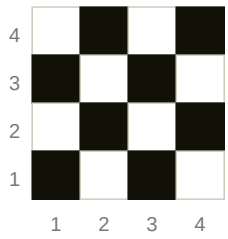


Figure 1 — Plain weave (1/1), two repeats shown. Every square alternates: no end is up for two picks in a row and no pick passes over two adjacent ends — the maximum possible interlacing, on a repeat of only 2 × 2.

5. The four records of a design

A weave drawn on paper is not yet a loom instruction. Four coordinated records convert it into one (Figure 2).

- **Design (weave plan).** The grid of Section 4: ends run, picks lifted.
- **Draft (drawing-in plan).** Placed above the design, one column per end: it states through which **heald shaft** each end is drawn. A **straight draft** draws ends 1, 2, 3, 4… on shafts 1, 2, 3, 4…; a **pointed draft** reverses at the turn (1, 2, 3, 4, 3, 2…); satin uses a **satin (distributed) draft**. The draft — not the design — decides how many shafts the loom needs: shafts must distinguish ends that lift differently.
- **Peg plan (lifting plan).** Placed beside the design, one row per pick and one column per shaft: a mark means that shaft is lifted for that pick. On a dobby it is the pegged lattice/chain; on a tappet loom, the cam order. Design + draft together determine the peg plan uniquely.
- **Denting plan.** How ends are distributed through the reed dents (e.g. 2 ends per dent), recorded as a separate line under the draft or stated with the reed count; it fixes the woven ends per unit width together with the reed. TE-3206 works a full denting example.

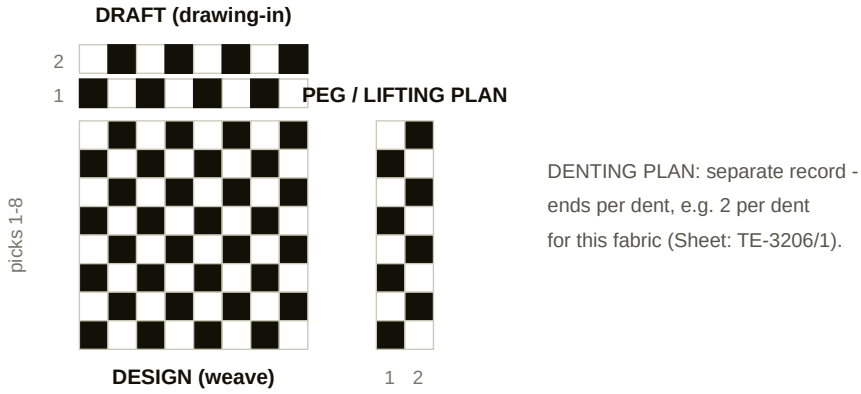


Figure 2 — The three graphical records for plain weave on a straight draft over 2 shafts: design (below), draft (above), peg plan (right). Because ends 1 and 3 lift alike, and ends 2 and 4 lift alike, only two shafts are needed although the repeat has four ends drawn here; the peg plan shows shaft 1 up on odd picks, shaft 2 on even picks.

READING CHECK — VERIFY FIGURE 2

End 1 is up on picks 1, 3, 5, 7 and end 2 on picks 2, 4, 6, 8 (Figure 1). The draft puts odd ends on shaft 1 and even ends on shaft 2. Therefore the peg plan must lift shaft 1 on odd picks and shaft 2 on even picks — exactly what Figure 2 shows. If any of the three records disagrees with the other two, the records — not the loom — are wrong.

6. From records to fabric: what the structure controls

The interlacing plan fixes, before weaving begins: the **maximum sett** that can be woven (interlacing frequency limits how closely threads can pack), the **crimp balance** between warp and weft (hence width and length shrinkage in finishing), the **face character** (warp-faced, weft-faced or balanced), **cover and weight** potential, and **service behaviour** — seam slippage, abrasion resistance, drape. The rest of this course is the systematic tour: fundamentals (Topics 2–4), derived and decorative structures (Topics 5–6), compound structures (Topics 7–8), then the parallel knitted world (Topics 9–12), with both turned into measurements and reproduction specifications in TE-3206.

Glossary

Warp / weft — lengthwise / crosswise yarn systems of a woven fabric. **End** — one warp thread. **Pick** — one inserted weft thread. **Shed** — the V-opening of separated warp ends through which the pick passes. **Repeat** — smallest ends × picks unit that tiles the design. **Float** — yarn segment passing over/under two or more opposing threads without interlacing. **Design / draft / peg plan / denting plan**

— the interlacing grid; the heald assignment of ends; the shaft-lifting order per pick; the distribution of ends through reed dents.

Key points

• A weave is nothing but the pattern of raised/lowered ends per pick; a filled square = end raised = warp on the face.
• Designs are read with pick 1 at the bottom. The repeat is quoted ends × picks.
• Three graphical records (design, draft, peg plan) plus the denting plan convert a pattern into a loom instruction; shafts follow the draft, and ends that lift identically may share a shaft.

Practice

1. A design occupies 6 ends × 4 picks before it repeats. State the repeat in standard form, and the minimum number of picks that must be drawn to show two full repeats vertically.
2. In a design, end 5 is raised on picks 2, 4 and 7 only within one repeat of 8 picks. On which picks does the weft show over end 5?
3. Four ends lift as follows: end A up on odd picks, end B up on even picks, end C exactly as A, end D exactly as B. What is the smallest draft (fewest shafts), and what does the peg plan look like?
4. Explain why a weave design alone, without draft and denting, cannot be sent to a loom as a complete instruction.

Answer key

1. Repeat = 6 × 4; two vertical repeats need 8 pick-rows drawn.
2. On all other picks of the repeat: 1, 3, 5, 6, 8 (the end is down, so the pick passes over it).
3. Two shafts: A and C on shaft 1, B and D on shaft 2. Peg plan: shaft 1 marked on odd picks, shaft 2 on even picks — the plain-weave lifting of Figure 2.
4. The design states interlacing but not how ends are grouped onto shafts (draft), nor the shaft-lifting mechanism order (peg plan — actually derivable from design + draft, so it must at least be checked), nor how the ends are spaced in width (denting/reed), which sets ends per cm and therefore the fabric itself.